Assignment 1 – LCC of Energy Measures
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1. Introduction

This report investigates the life-cycle cost (LCC) analysis of a church in Ängelholm, Sweden, named Rönnekyrkan. LCC is a tool used to determine the overall financial performance of a product, in this case, a building component. LCC covers costs starting from the initial investment, to the operation and maintenance of the product throughout its life-cycle. This method also considers long-term financial implications, such as changes in future prices and interest rates. This way, it is possible to have a feasible economic strategy and reduce the overall investment cost for a product. Therefore, it is crucial to implement LCC analysis to ensure a balance between more informed design decisions and optimised investments.

2. Method

The Rönnekyrkan is a church built in the 1930s with a total built-up area of 420 m. It has three floors. The ground floor has a foyer, chapel, parlour, kitchen, office room, toilets and stair room. The first floor is a residential unit for the family of the caretaker of the church, and the basement is used to conduct activities, a laundry room, and store equipment used in the church.

For this project, two energy-saving measures were proposed for the church. LCC analysis, as well as sensitivity analysis, was performed for each case using Microsoft Excel calculations. A site visit was conducted to observe the properties of the building, and some assumptions had to be made in the process. The chosen two measures were changing the hot water system and changing the front door of the building.

2.1. LCC Calculation

LCC calculation was done using the formulas provided in the course [9]. *Table 1* shows the input used in these calculations. The heating and electricity prices were taken from the average of 2024 bills.

Table 1: LCC Properties

rn (nominal interest rate) [6]	2%	rf (inflation) [6]	3%
rci (nominal price increase)	2.5%	rpi (real price increase)	-0.49%
rr (real interest rate)	-0.97%	rdi (discount interest rate)	-0.49%
Heating price (SEK/kWh) [10]	1.38	Electricity price (SEK/kWh) [10]	1.60
Water price (SEK/Litr) [12]	0.039		

Utilising these parameters, Formula 1 was used to calculate the LCC of both measures. The initial cost and expenditure of the current value for each measure will be explained in the next sections.

$$\mathrm{PVi} = \frac{\mathrm{Ci}}{(1+\mathrm{rdi})^{n}} \tag{1}$$

2.2. Measure 1: Hot Water

Evaluating the economic viability of two water-saving renovation options against a Base Case of the building with 25 occupants begins with establishing baseline measurements derived from user and occupancy schedules. The methodology follows a systematic process where general parameters were first defined, including the number of occupants (25 PAX), annual workdays (34.7 days), and utility prices (water: 0.039 SEK/L, electricity: 1.38 SEK/kWh).

Table 2: Differences between options

	Category	Qty	Unit Price (SEK)	Line Total (SEK)
Op1	Restroom sensor taps [13]	2	2000	4000
	Kitchen sensor taps [14]	1	2500	2500
	CapEx (SEK)			**6500**

	Category	Qty	Unit Price (SEK)	Line Total (SEK)
Op2	Electric Water Heater (30L) [5]	1	1400	1400
	Fittings & Materials (Lumpsum)	1	800	800
	Restroom sensor taps [13]	2	2000	4000
	Kitchen sensor taps [14]	1	2500	2500
	Labour (Plumber based on 6 hours of work at 700 SEK/hour)	6	700	4200
	CapEx (SEK)			**12900**

Two options were analysed: Option 1 installed efficient sensor taps in kitchens and toilets, while Option 2 added an electric water heater (*Table 2)*. Water consumption was calculated for each scenario, with the Base Case using standard flow rates (9 L/min for toilets, 0.333 L/min for restroom taps, and 8 L/min for kitchen taps). Both renovation options implement sensor taps with reduced flow rates (restroom: 0.133 L/min, kitchen: 5 L/min). Hot water demand was estimated at 40% of total consumption, heated to 55°C, assuming four-day weekly operation.

Capital expenditures were sourced from detailed cost sheets, showing Op1 requires 6,500 SEK for sensor taps while Op2 requires 12,900 SEK for taps, water heater, fittings, and labour. The Base Case has no initial investment. Operational expenditures combine water supply costs and, for Op2, electricity for water heating calculated using the sensible heat transfer formula at 75% efficiency.

Table 3: Sensitivity Analysis on Discount Rates and Energy Prices.

Interest	Value	%
Rdi (discount interest rate)	-0.0049	-0.4878
Rn (nominal interest rate)	0.02	2
Rf (inflation)	0.03	3
Rpi (real price increase)	-0.0049	-0.4854

Rci (nominal price increase)	0.025	2.5
Rr (real interest rate)	-0.0097	-0.97

The analysis was conducted over a 20-year lifetime, with future costs discounted using a real rate of -0.97% derived from 2% nominal interest and 3% inflation. Total LCC represents the sum of CapEx and the present value of all future costs, complemented by sensitivity analysis on discount rates and energy prices. The comparison of options is presented in *Table 3*, with detailed hot water usage estimation shown in *Table 4*.

Table 4: Total Hot Water Usage

Hot Water System									
Hot Water Consumption	**Case**	**Ratio**	**Base Case**	**Op1 & Op2**	**Hw_Base Case**	**Hw_Op1**	**Hw_Op2**		
Toilets	Restroom tap	0.4	1732	138	693	55	Electric Water Heater	Electric Water Heater (30L)	30
Kitchen	Kitchen uses per person per day	0.4	3467	724	1387	289		Working Day at (5days in a week)	260
Total Water Consumption - Ltrs					**2079**	**345**		**Total heating hours**	**7800**

Using these, the total energy input was found using Formulas 2, 3 and 4.

$Q = m \times Cp \times \Delta T$ (2)
Electrical Energy Input = Thermal Energy (kWh) / Efficiency (3)
Cost = Electrical Energy Input × Electricity Price (4)

2.3. Measure 2: Front Door

The existing door of the church was considered Door 1. In addition to the existing door, the implementation of a door closer was considered as Door 2. As the last alternative, changing the door to an automatic one was considered as Door 3. The heat losses from both conduction and door openings were calculated using Formulas 5, 6, 7, 8, 9, and 10 [3]. In the calculations, climate data from Halmstad were used [11]. *Table 5* shows the assumptions made about each case. It was also assumed that the church would be in use 4 days a week for various activities. Then, the LCC analysis was conducted for a 20-year lifetime, using the parameters shown in the LCC Calculation section. As for a sensitivity analysis, different r_{di} values were tested along with higher heating prices.

$$E_{conduction} = U \times A \times Gh/1000 \tag{5}$$

$$q = Cd\frac{B}{3}\sqrt{g' \times H^3} \tag{6}$$

$$g' = 2 \times g\,\frac{Ti\text{-}To}{Ti + To} \tag{7}$$

$$Cd = 0.4 + 0.0045 \times |Ti\text{-}To| \quad (8)$$

$$Q = \rho \times q \times cp \times (T_i - T_o) \quad (9)$$

$$E_{opening} = Q \times t \quad (10)$$

Table 5: Assumptions for different doors

Cases	U-Value (W/m²K)	The amount of time the door is open for in a day (s)	Cost of purchase, workmanship cost included as 40% (SEK)	Cost of maintenance, annual (SEK)	Cost of operation (electricity), annual (SEK)
Door 1	2.5	3600	0	0	0
Door 2	2.5	1500	3200 [4]	500	64
Door 3	1.4 [8]	750	50000 [7]	2800	120

The calculated heat losses and the operational electricity were taken as Ci in the calculation, while the purchase cost, along with the workmanship, was taken as the initial cost. Maintenance cost was added to each year.

3. Results

3.1. Measure 1: Hot Water

The analysis reveals significant differences in performance between the scenarios. Op1 reduces total annual water consumption from 12,294 litres (Base Case) to 3,299 litres, a 73% reduction. This translates to an annual water cost saving of 351 SEK (Base: 479 SEK, Op1: 128 SEK) (*Table A).*

Op2 achieves the same water savings as Op1. However, its operation requires electricity for water heating, resulting in an annual energy cost of 1,053 SEK. Consequently, its total annual utility cost (water + electricity) is 1,181 SEK (*Table B and Table C).*

The Life Cycle Cost analysis reflects these operational differences over 20 years:

- *Base Case LCC: 13,701 SEK (no initial investment)*
- *Op1 LCC: 10,772 SEK (6,500 SEK investment)*
- *Op2 LCC: 39,209 SEK (15,400 SEK investment)*

The payback period for Op1 compared to the Base Case is approximately 34.2 years, which exceeds the analysis period, while Op2 does not achieve payback.

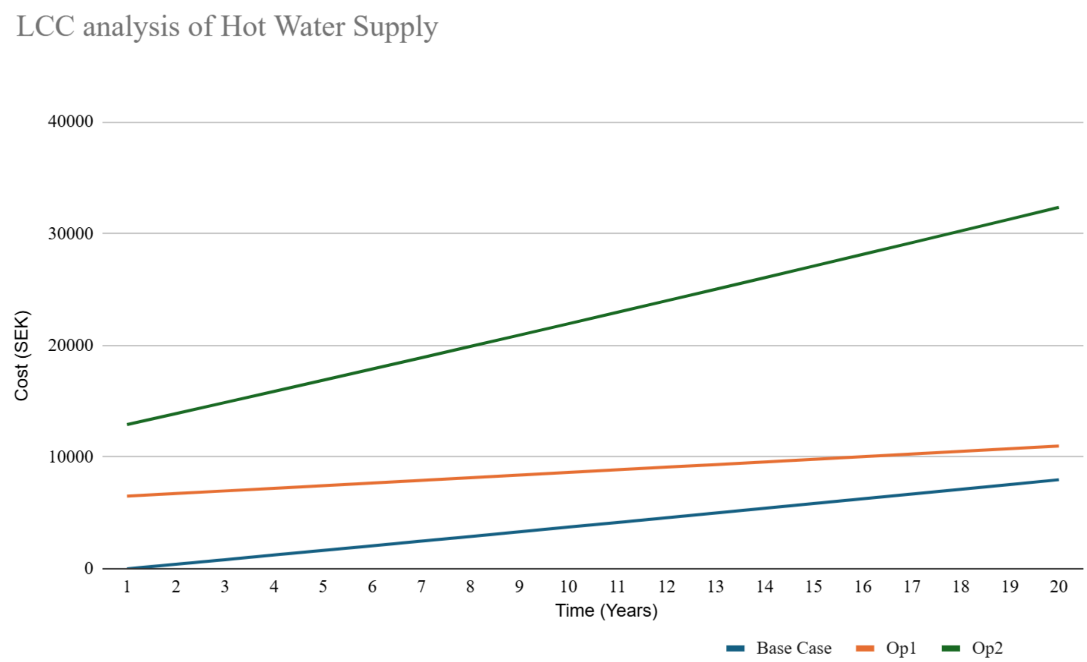

Figure 1: LCC analysis of hot water supply

3.2. Measure 2: Front Door

The annual heating losses of Door 3 were the lowest, as seen in Table 6. Figure 2 shows the LCC analysis result of each case. The payback time for Door 2 was calculated as 0.19 years, while for Door 3, it was 2.38 years. It was seen that Door 2 reduces life-cycle costs by around 54% compared to Door 1, while Door 3 achieves an even higher reduction with 61%. At the end of the 20-year life span, Door 3 is 15% more cost-efficient compared to Door 2.

Table 6: Heating losses from different doors

Cases	Heating loss from conduction annually (kWh)	Heating loss from door openings annually (kWh)	Total heating loss annually (kWh)
Door 1	817.04	1294.62	2111.66
Door 2	457.54	539.43	1356.46
Door 3	457.54	269.71	727.25

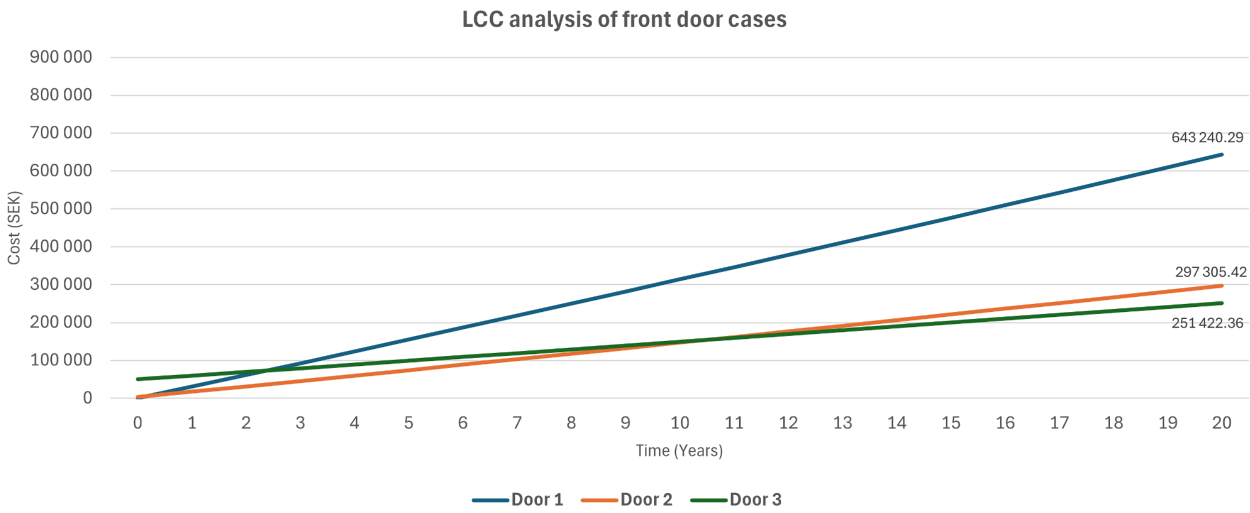

Figure 2: LCC analysis of front door cases

4. Discussion

4.1. Measure 1: Hot Water

Sensitivity analysis shows these results are robust to variations in discount rates and energy prices within reasonable ranges.

Therefore, based on this LCC analysis, Op1 is the most economically favourable option despite its long payback period, while Op2 is not economically viable due to its high combined investment and operational costs.

The analysis compared two renovation options against the existing Base Case. Option 1 installed sensor taps in restrooms and the kitchen, reducing flow rates to achieve a 73% reduction in annual water consumption (from 12,294 L to 3,299 L). This required a 6,500 SEK investment but lowered maintenance costs by 83%. However, its payback period is 34 years.

Option 2 added an electric water heater to the sensor taps, costing 12,900 SEK. While it achieved the same water savings, the added electricity demand for heating made it economically unviable, with significantly higher operational costs.

Despite the long payback period, Option 1 is recommended for its substantial environmental benefits and lower operating costs, making it the superior choice for long-term sustainability.

4.2. Measure 2: Front Door

Front door results show that the most cost-effective option is Door 3, even though it has a slightly longer payback period. This is due to the fact that Door 3 has a high investment cost, while the operational costs are lower than in other cases. Therefore, a sensitivity analysis was conducted on this last option. Two scenarios were tried: changing the r_{di} value and changing the heating price. It was seen that having a higher r_{di}, 0.49% instead of -0.49%, gives a slightly longer payback period for Door 3, increasing from 2.38 to 2.42. This change is negligible.

The second scenario, where the heating price is increased from 1.38 SEK/kWh to 1.80 SEK/kWh, had a larger change. The payback time for Door 3 decreases from 2.38 to 1.77 in this case. These results show that in the case of energy prices increasing in the future, investing in Door 3 can be even more cost-effective, due to its low operating costs.

5. Conclusion

This analysis reveals a clear trade-off between sustainability and economic feasibility. The hot water system renovation (Option 1) delivers significant environmental benefits in water reduction, but its longer payback period and higher initial investment present long-term economic challenges. In contrast, the front door replacement offers immediate financial viability with a shorter payback, making it the superior standalone investment.

The front door case shows that the case with lower investment is easier to achieve payback, while in the long-term results, it is seen that the option with a higher initial investment is more cost-effective at the end of the life-cycle.

The final decision should balance the church's budget constraints against sustainability goals, with the door replacement (Door 3) providing the most compelling economic case while for the hot water usage, Option 1 remains the better environmental solution.

6. References

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7. Abbreviations

Short Form	Abbreviations
LCC	Life Cycle Cost
LCA	Life Cycle Assessment
NPV	Net Present Value
PV	Present Value
CapEx	Capital Expenditure (initial investment)
OpEx	Operating Expenditure (annual)
N	Analysis lifetime (years)
r	Discount rate (per year)
ppd	people per day

8. Appendix

Table A: Total Water Consumption

Total Water Consumption			
General Water Consumption	**Case**	**Base Case**	**Op1 & Op2**
Toilets	WC Usage	7800	7800
	Restroom tap	1732	138
Kitchen	Kitchen uses per person per day	3467	724
Total Water Consumption - Ltrs		12998	8662
Price of water		**507**	**338**

Table B: Operation Cost

Total Operational Costs	
Base Case (Water + Heating)	634.83
Op1 (Water + Fixtures + Heating)	460.49
Op2 (Water + Fixtures + Electric heating)	1108.34

Table C: Operation Cost

Sensible heat transfer	Hw_Base Case	Hw_Op1	Hw_Op2	Units
Mass(M)	2079	345	7800	Kg
Temp_Initial	55	55	55	°C
GTemp_Final	8	8	8	°C
ΔT = T_final - T_initial	47	47	47	°C
Cp (specific heat capacity) of water	4184	4184	4184	J/kg°C

Calculations				
A) In kJ	408891497.4	67803181.61	1533854400	kJ
B) In Kilowatt-hours (kWh)	113.5809715	18.83421711	426.0706667	kWh
Efficiency (η)				
0.75	**151.4412953**	**25.11228949**	**568.0942222**	**kWh**
Electricity Cost	**208.9889876**	**34.65495949**	**783.9700267**	**SEK**